MIT 8103 Advanced Database Systems: supporting evidence

Student: Aregbe Abiodun Micheal · **Student ID:** 301915034 · **Matric:** 2026/A/MIT/0502
Case study: LagosLink Deliveries (simulated) · **GitHub:** github.com/AregbeAMicheal/advanced-database-portfolio

Key screenshots from my own runs, two or three per portfolio, to accompany the final documentation. Every screenshot, script and README is in the matching folder of the GitHub repository.

Portfolio 1: Database design and modelling (PostgreSQL 18, pgAdmin 4)

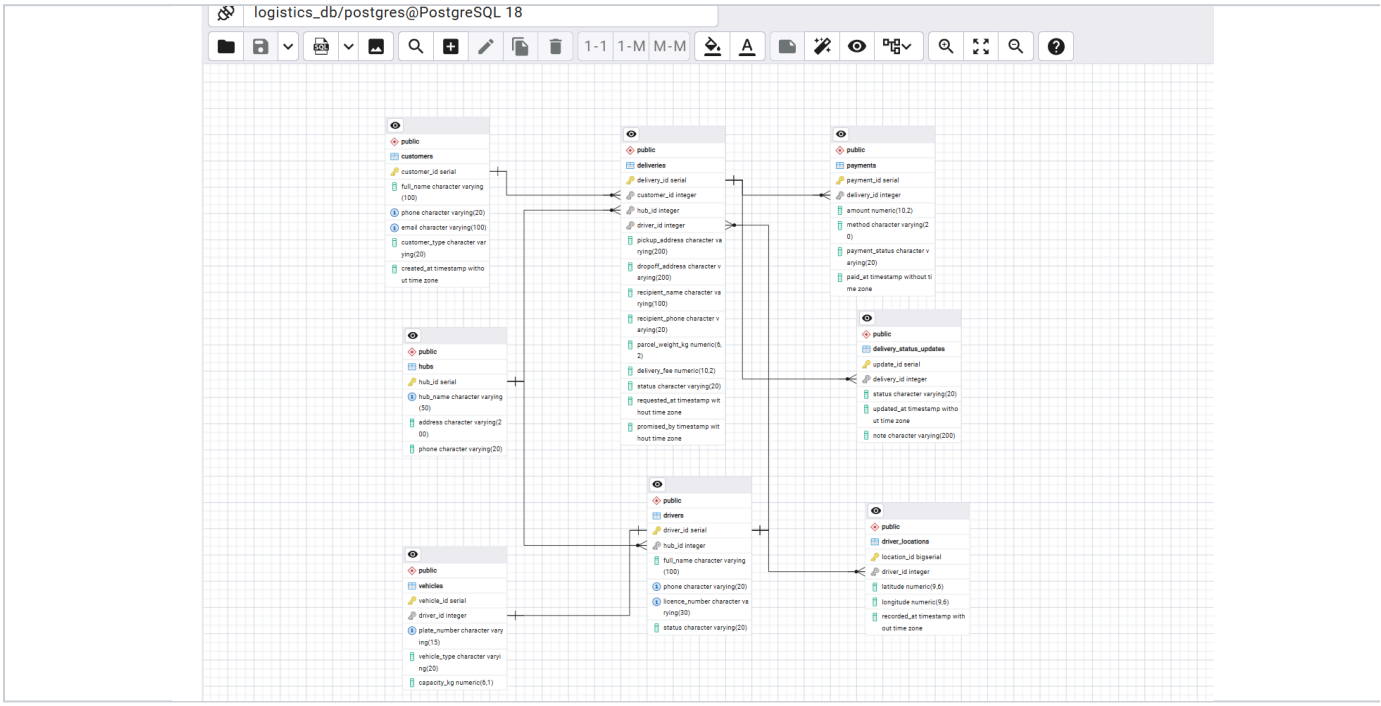


Figure 1. ER diagram generated by pgAdmin's ERD Tool from the 8 LagosLink tables, showing primary keys, foreign keys and one-to-many / one-to-one relationships.

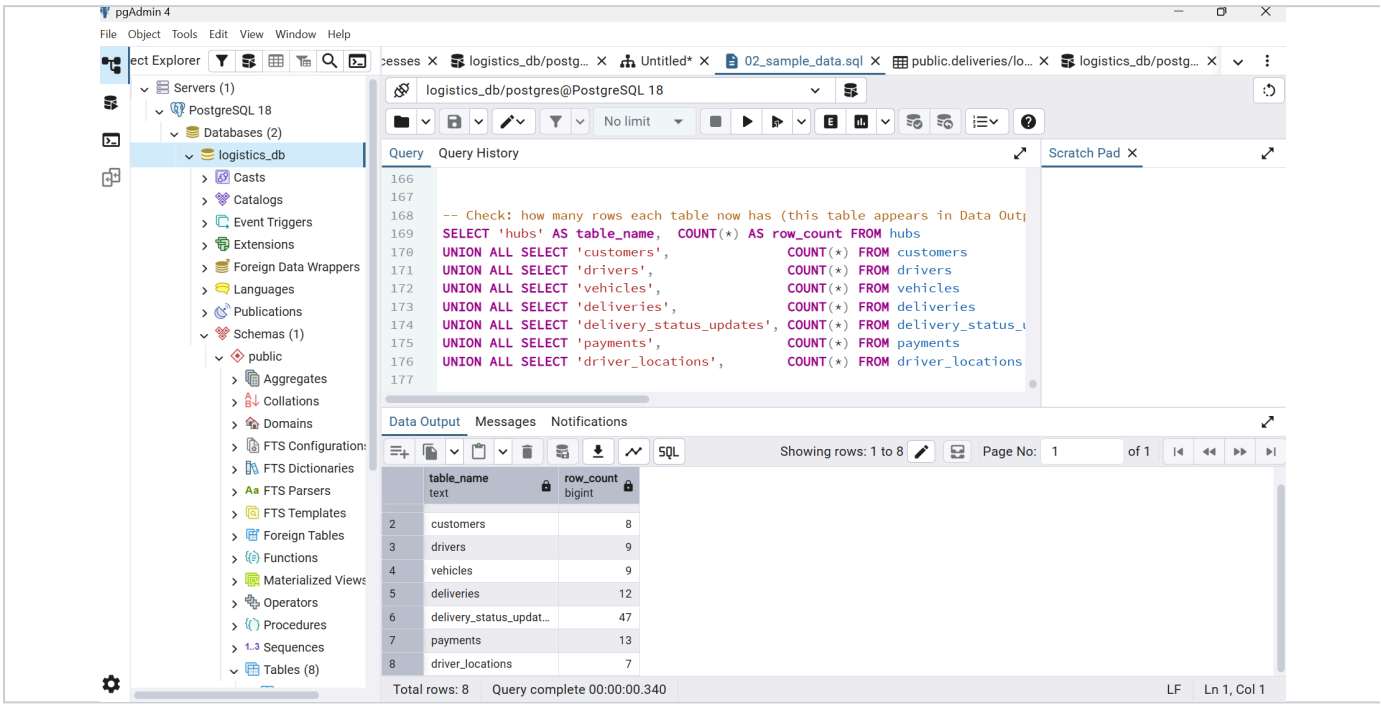


Figure 2. Sample data loaded into all 8 tables (row count per table).

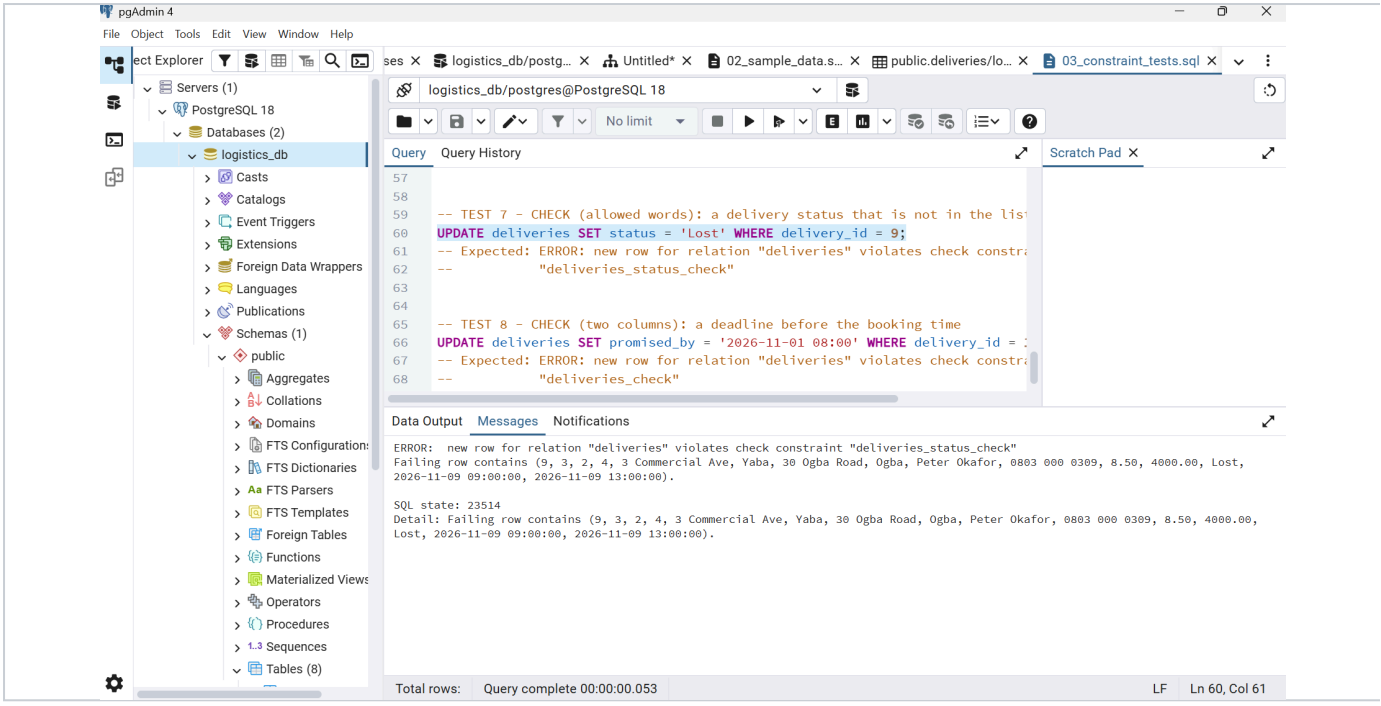


Figure 3. Constraint test 7: setting a delivery's status to 'Lost' is refused by the CHECK constraint `deliveries_status_check` (one of 8 tests, all refused).

Portfolio 2: Query processing and optimisation

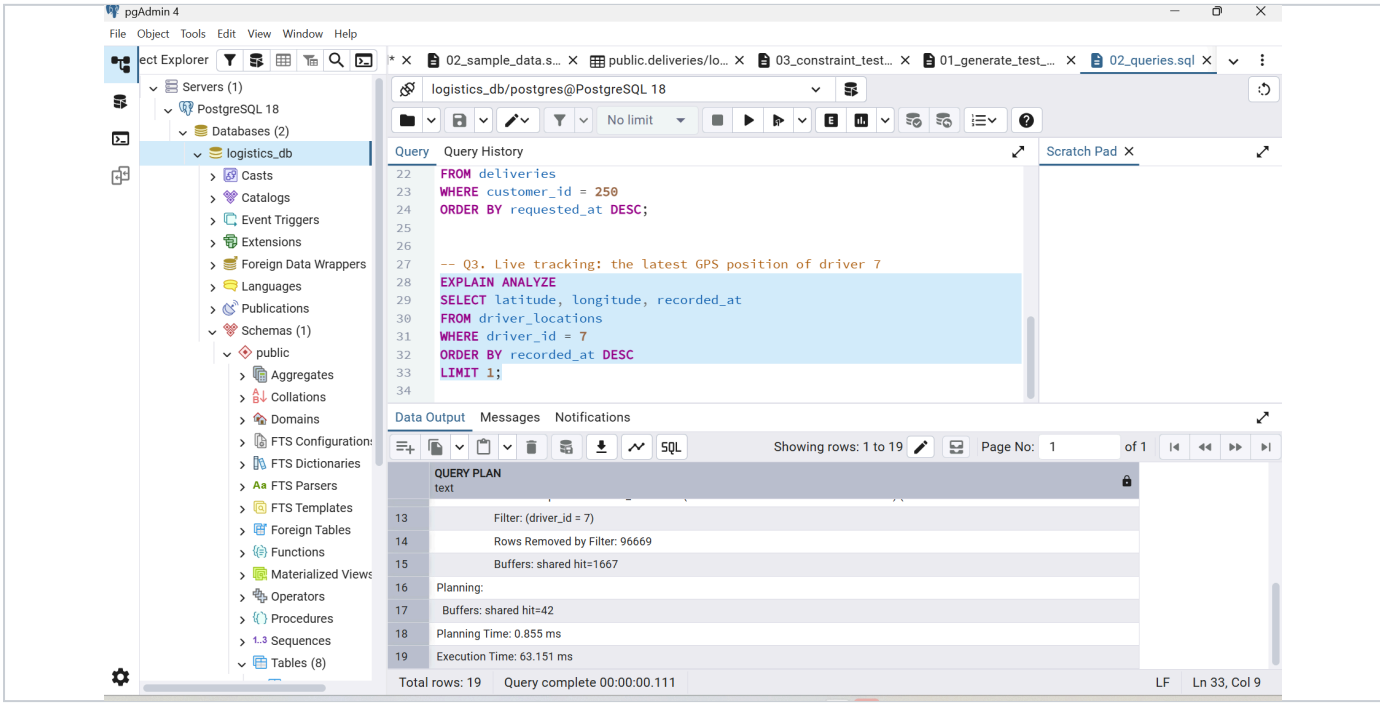


Figure 4. Q3 (a driver's latest GPS point) before indexing: sequential scan, 96,669 rows removed by the filter, 1,667 pages read, execution time 63.151 ms.

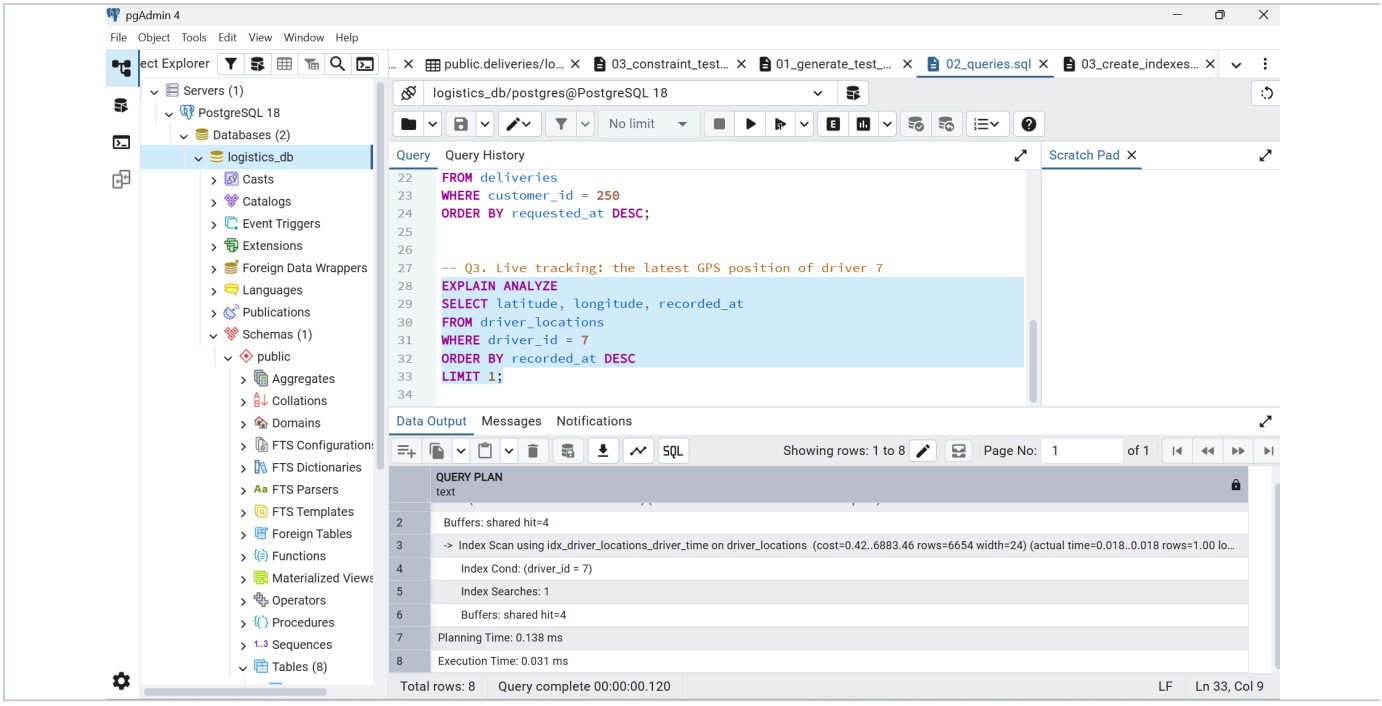


Figure 5. Q3 after creating the composite index `idx_driver_locations_driver_time`: Index Scan, 4 pages read, execution time 0.031 ms (about 2,037 times faster).

Portfolio 3: Transactions and concurrency

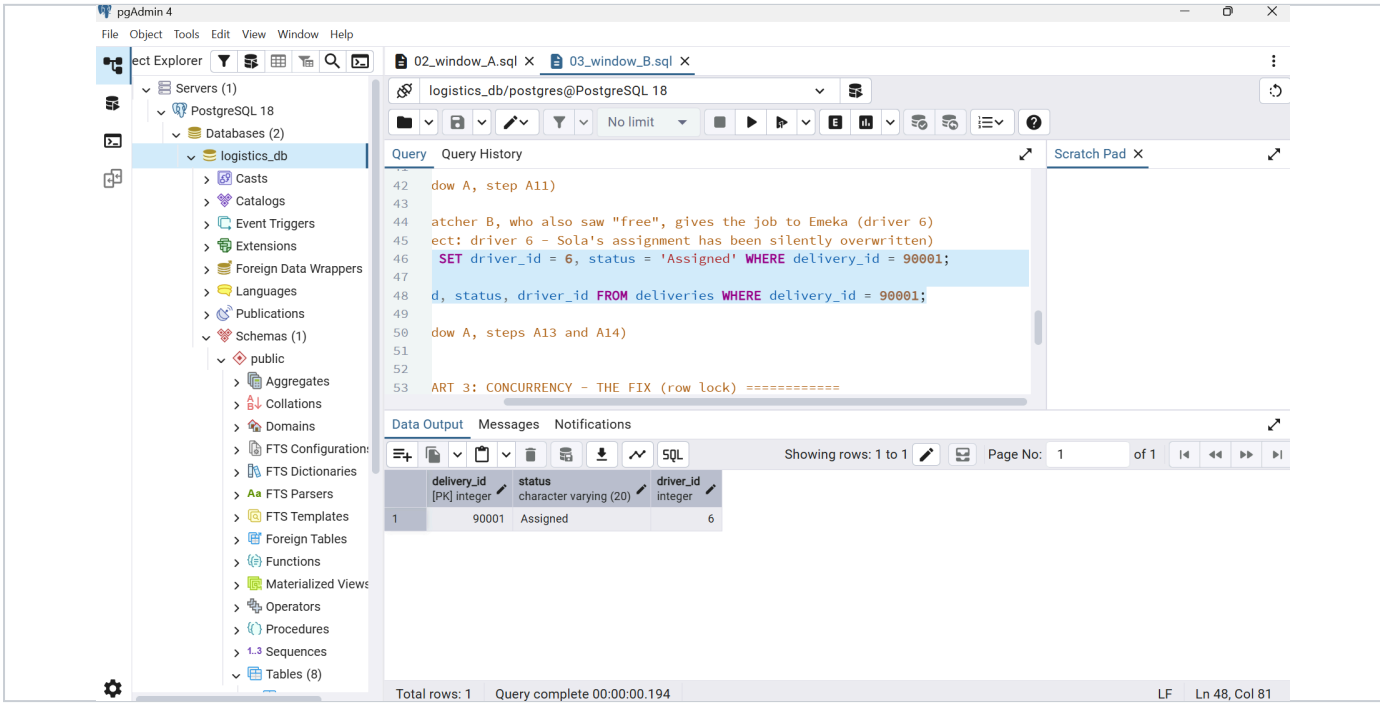


Figure 6. Lost update: Dispatcher B's assignment (driver 6) silently overwrote Dispatcher A's committed assignment (driver 5) of delivery 90001, with no error.

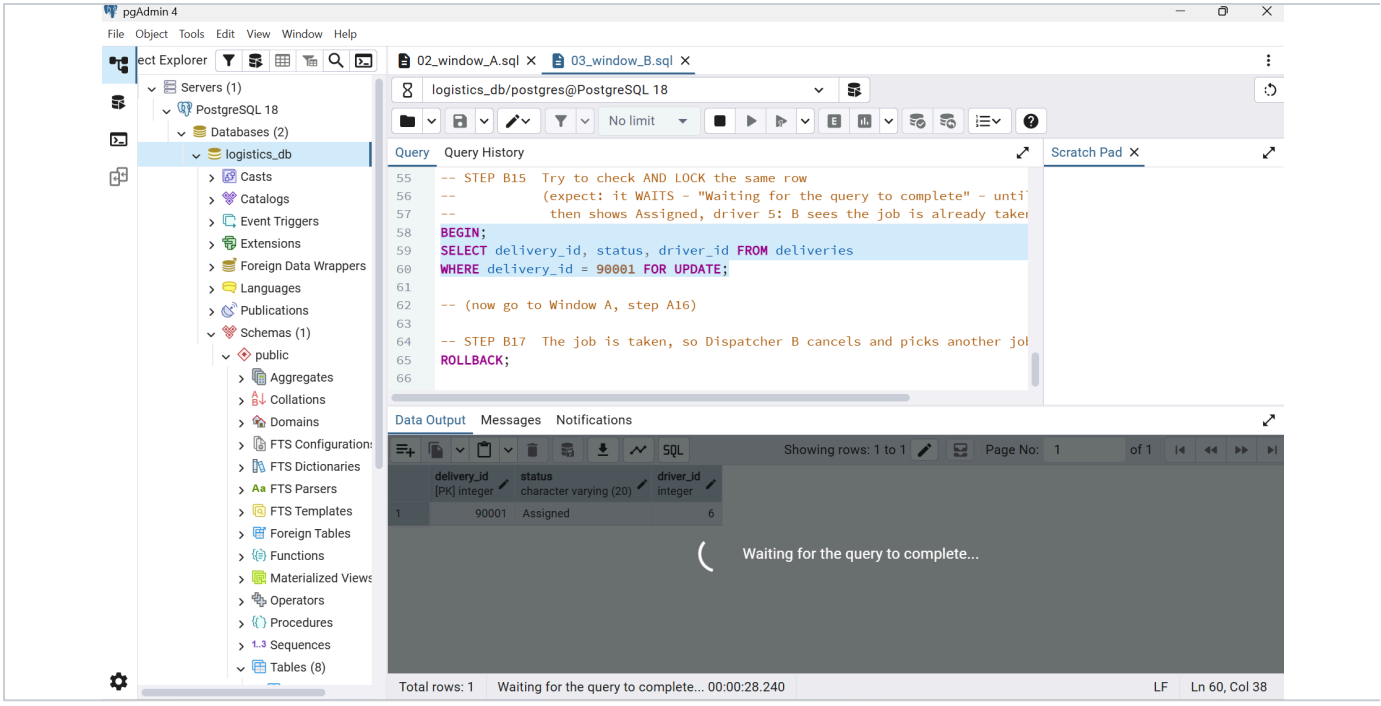


Figure 7. The fix: Dispatcher B's `SELECT ... FOR UPDATE` waits ("Waiting for the query to complete") while Dispatcher A holds the row lock. After A committed, B saw the job was taken and rolled back.

Portfolio 4: NoSQL document model (MongoDB Atlas, Compass mongosh)

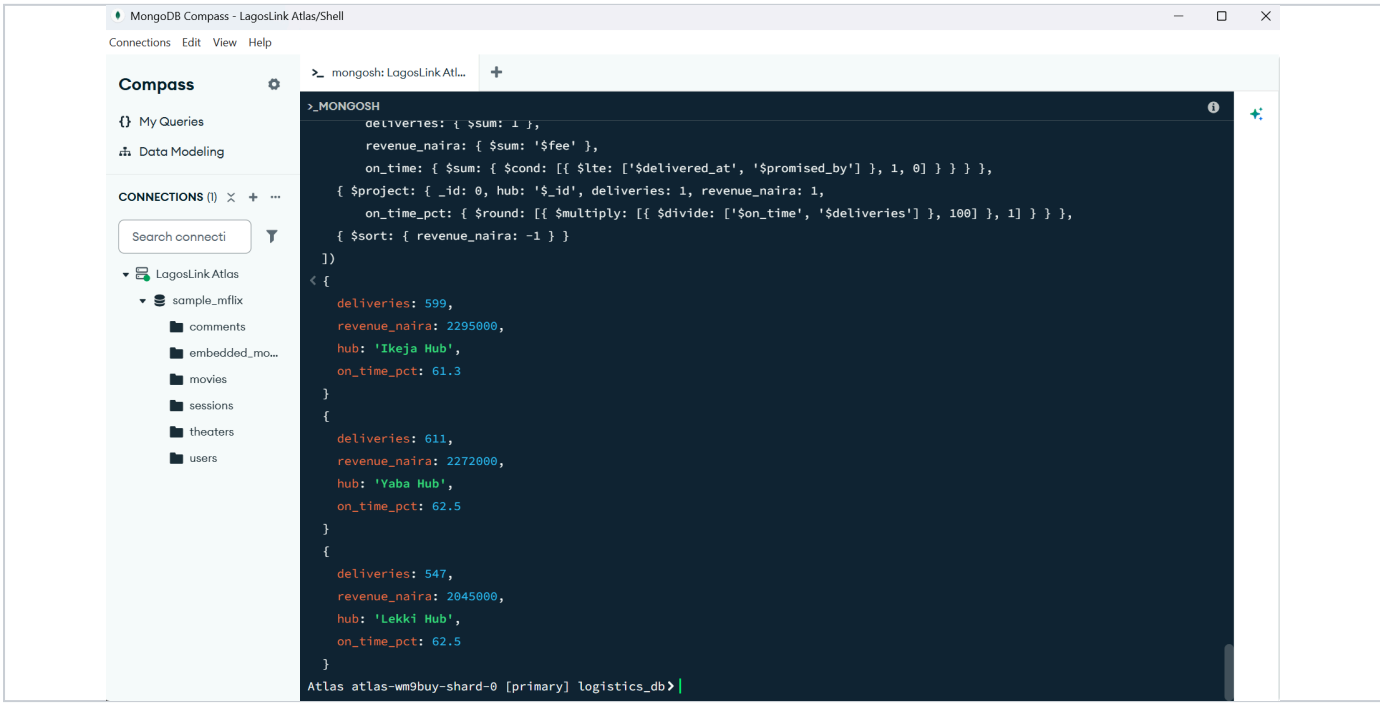


Figure 8. Q3 aggregation pipeline (hub report): Ikeja 599 deliveries / ₦2,295,000 / 61.3% on time; Yaba 611 / ₦2,272,000 / 62.5%; Lekki 547 / ₦2,045,000 / 62.5%.

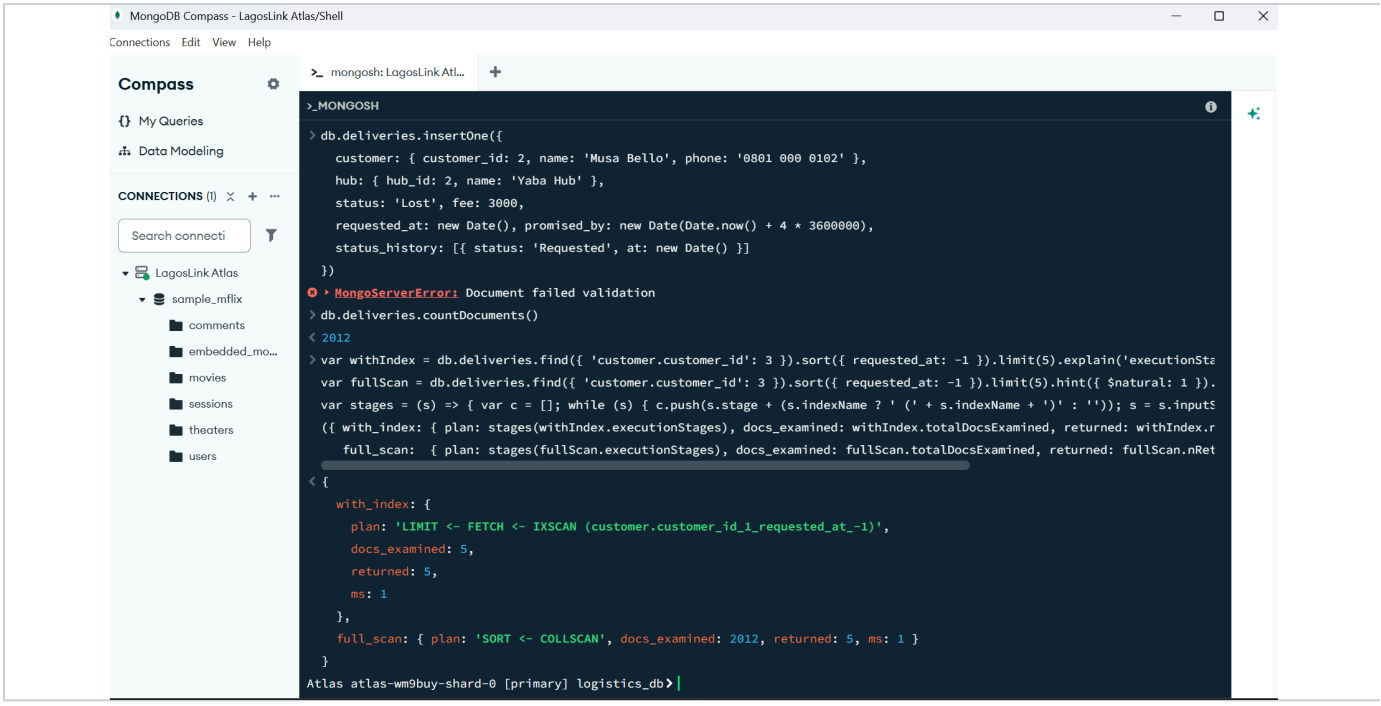


Figure 9. Q6: a delivery with status 'Lost' is rejected by the JSON Schema validator ("Document failed validation"; the count stays 2,012). Q7: with the index the query examined 5 documents (IXSCAN); forced to scan, it examined 2,012 (COLLSCAN).

Portfolio 5: Distributed and cloud databases

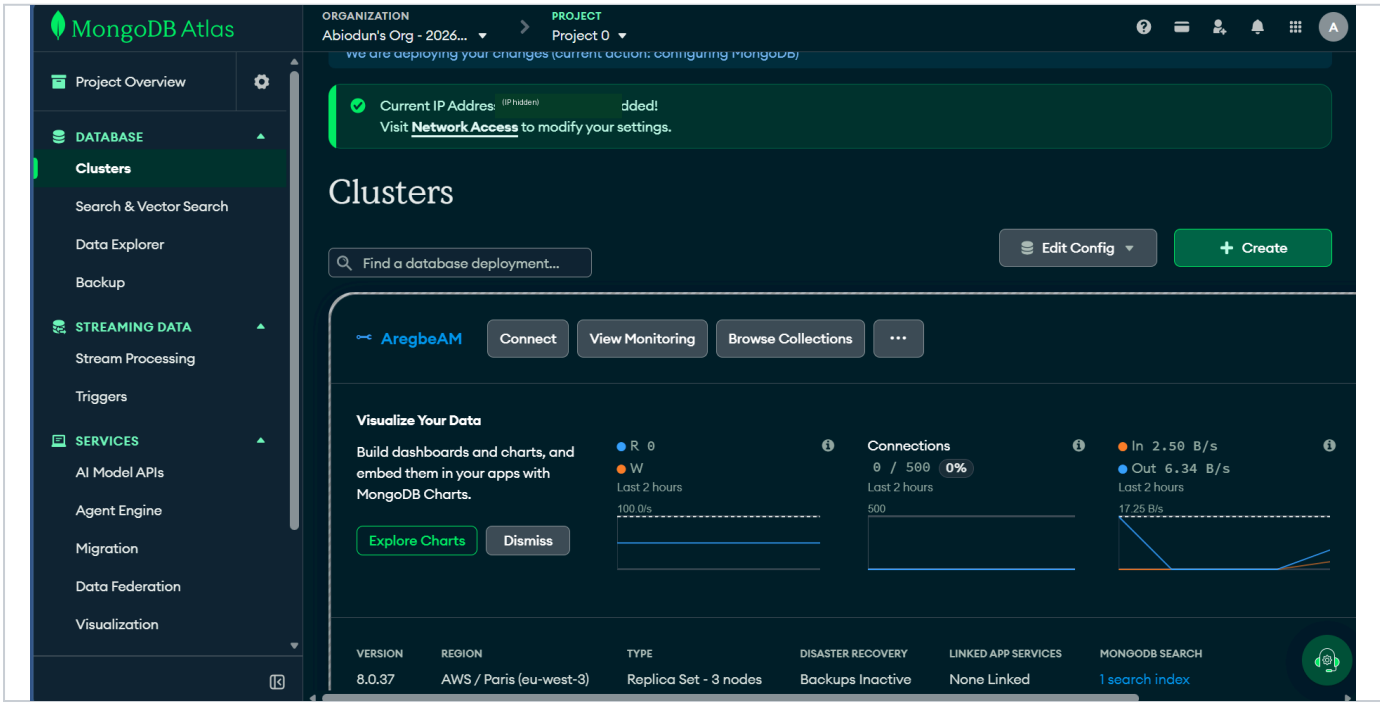


Figure 10. The Atlas cluster `AregbeAM`: AWS / Paris (eu-west-3), Replica Set - 3 nodes, MongoDB 8.0 (my IP address hidden).

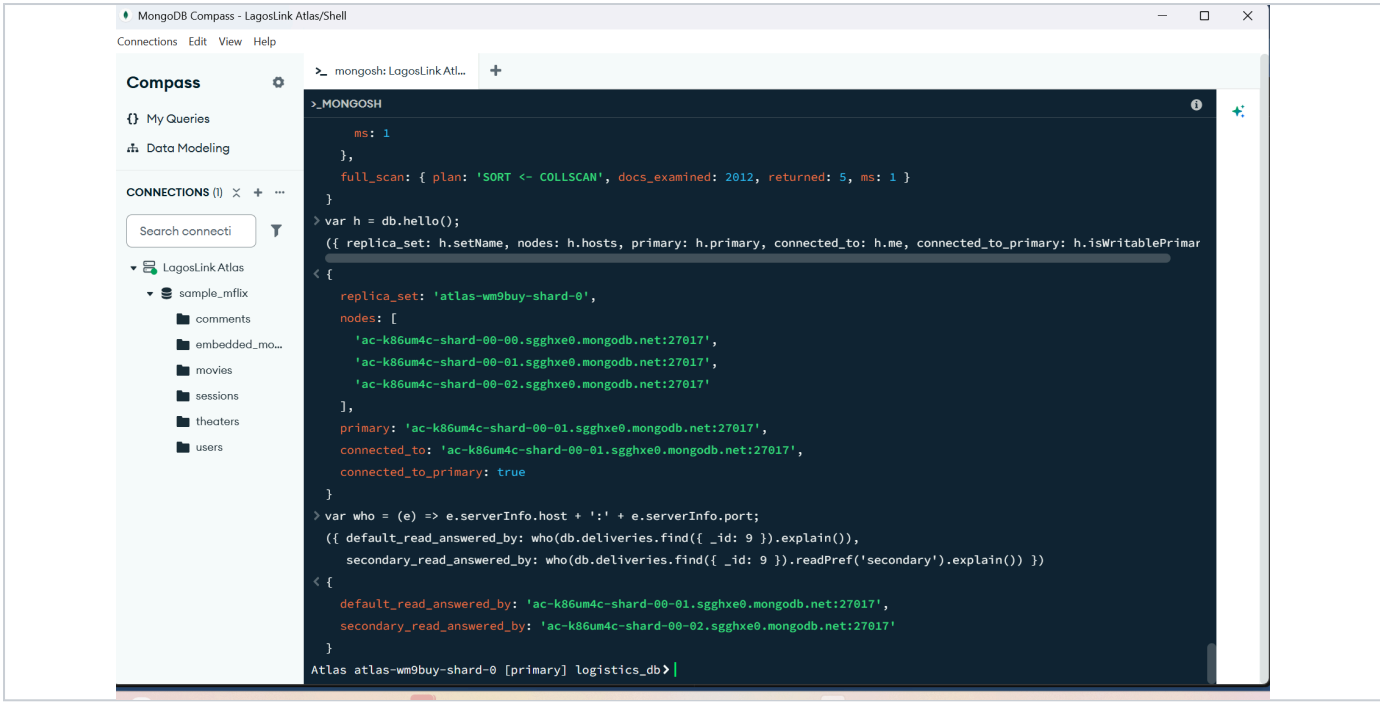


Figure 11. `db.hello()`: replica set `atlas-wm9buy-shard-0` with 3 nodes and primary `shard-00-01`. The same read was answered by the primary by default and by secondary `shard-00-02` with `readPref('secondary')`.

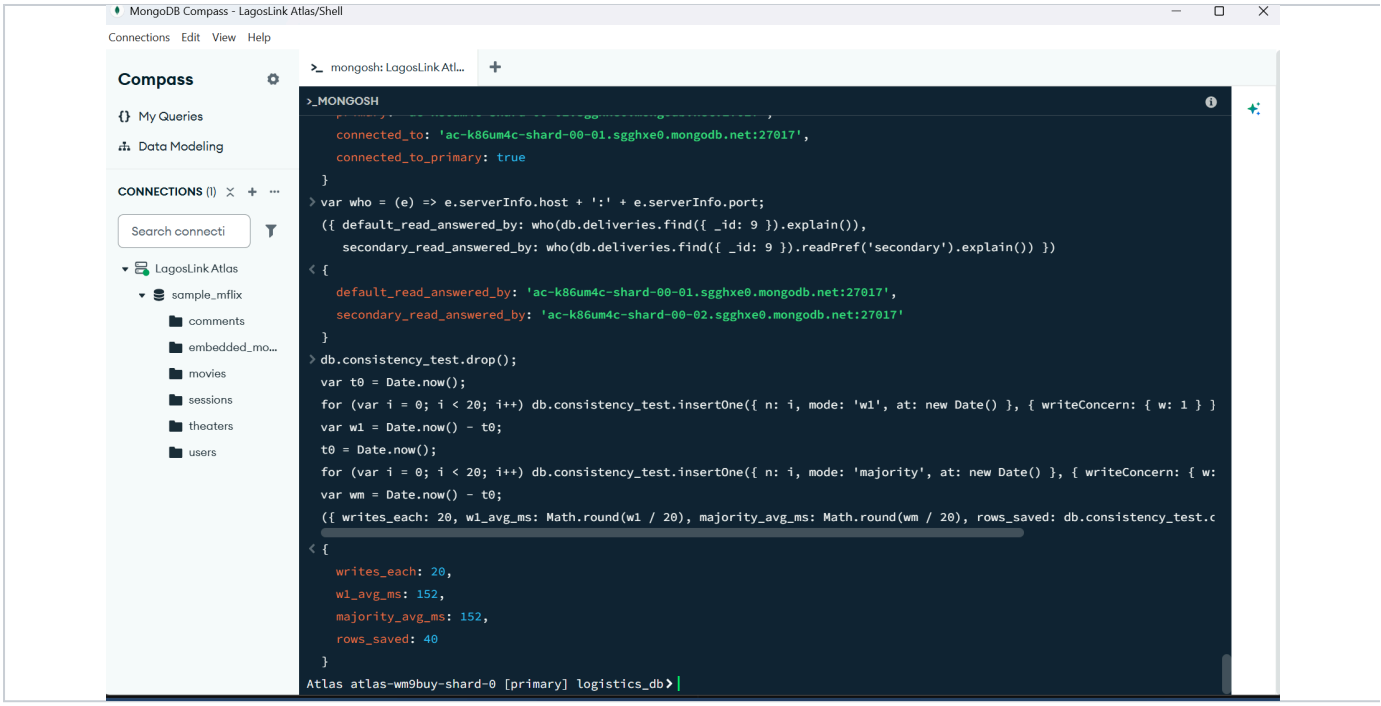


Figure 12. Write concern: 20 writes each averaged 152 ms with `w: 1` and 152 ms with `w: "majority"` (40 records saved); the network trip from Lagos to Paris dominates.